GenSmart Optimization Report

(Tool Version Beta 1.0)

Job ID:	20260309084636794045
Date:	2026-03-09 20:46:37
Gene Name:	NC
Expression Host Organism:	Komagataella pastoris (Pichia pastoris)
Sequence Type:	DNA
Size:	2766bp
Excluded enzyme sites:	EcoRI [GAATTC]; NotI [GCGGCCGC]
Kept enzyme sites:	[]

Original Sequence (Original Sequence Length: 2766bp, GC%:60.52%):

TTCTTCCCCTTCATCTCGAGAGGGAAAGAACTCCTTTGGGGAAAGCCTGAGGAGTCTCGTGTCTCTAGCGTCTTGGAGGAAAGCAAGCGCCTGGTGGACACCGCCATGTACG
CCACGATGCAGAGAAACCTCAAGAAAAGAGGAATCCTTTCTCCAGCTCAGCTTCTGTCTTTTTCCAAACTTCCTGAGCCAACAAGCGGAGTGATTGCCCGAGCAGCAGAGATA
ATGGAAACATCAATACAAGCGATGAAAAGAAAAGTCAACCTGAAAACTCAACAATCACAGCATCCAACGGATGCTTTATCAGAAGATCTGCTGAGCATCATTGCAAACATGTCT
GGATGTCTCCCTTACATGCTGCCCCCAAAATGCCCAAACACTTGCCTGGCGAACAAATACAGGCCCATCACAGGAGCTTGCAACAACAGAGACCACCCCAGATGGGGCGCCT
CCAACACGGCCCTGGCACGATGGCTCCCTCCAGTCTATGAGGACGGCTTCAGTCAGCCCCGAGGCTGGAACCCCGGCTTCTTGTACAACGGGTTCCCACTGCCCCCGGTCC
GGGAGGTGACAAGACATGTCATTCAAGTTTCAAATGAGGTTGTCACAGATGATGACCGCTATTCTGACCTCCTGATGGCATGGGGACAATACATCGACCACGACATCGCGTTC
ACACCACAGAGCACCAGCAAAGCTGCCTTCGGGGGAGGGGCTGACTGCCAGATGACTTGTGAGAACCAAAACCCATGTTTTCCCATACAACTCCCGGAGGAGGCCCGGCCG
GCCGCGGGCACCGCCTGTCTGCCCTTCTACCGCTCTTCGGCCGCCTGCGGCACCGGGGACCAAGGCGCGCTCTTTGGGAACCTGTCCACGGCCAACCCGCGGCAGCAGAT
GAACGGGTTGACCTCGTTCCTGGACGCGTCCACCGTGTATGGCAGCTCCCCGGCCCTAGAGAGGCAGCTGCGGAACTGGACCAGTGCCGAAGGGCTGCTCCGCGTCCACG
CGCGCCTCCGGGACTCCGGCCGCGCCTACCTGCCCTTCGTGCCGCCACGCGCGCCTGCGGCCTGTGCGCCCGAGCCCGGCATCCCCGGAGAGACCCGCGGGCCCTGCTT
CCTGGCCGGAGACGGCCGCGCCAGCGAGGTCCCCTCCCTGACGGCACTGCACACGCTGTGGCTGCGCGAGCACAACCGCCTGGCCGCGGCGCTCAAGGCCCTCAATGCG
CACTGGAGCGCGGACGCCGTGTACCAGGAGGCGCGCAAGGTCGTGGGCGCTCTGCACCAGATCATCACCCTGAGGGATTACATCCCCAGGATCCTGGGACCCGAGGCCTT
CCAGCAGTACGTGGGTCCCTATGAAGGCTATGACTCCACCGCCAACCCCACTGTGTCCAACGTGTTCTCCACAGCCGCCTTCCGCTTCGGCCATGCCACGATCCACCCGCTG
GTGAGGAGGCTGGACGCCAGCTTCCAGGAGCACCCCGACCTGCCCGGGCTGTGGCTGCACCAGGCTTTCTTCAGCCCATGGACATTACTCCGTGGAGGTGGTTTGGACCCA
CTAATACGAGGCCTTCTTGCAAGACCAGCCAAACTGCAGGTGCAGGATCAGCTGATGAACGAGGAGCTGACGGAAAGGCTCTTTGTGCTGTCCAATTCCAGCACCTTGGATC
TGGCGTCCATCAACCTGCAGAGGGGCCGGGACCACGGGCTGCCAGGTTACAATGAGTGGAGGGAGTTCTGCGGCCTGCCTCGCCTGGAGACCCCCGCTGACCTGAGCACA
GCCATCGCCAGCAGGAGCGTGGCCGACAAGATCCTGGACTTGTACAAGCATCCTGACAACATCGATGTCTGGCTGGGAGGCTTAGCTGAAAACTTCCTCCCCAGGGCTCGG
ACAGGGCCCCTGTTTGCCTGTCTCATTGGGAAGCAGATGAAGGCTCTGCGGGACGGTGACTGGTTTTGGTGGGAGAACAGCCACGTCTTCACGGATGCACAGAGGCGTGAG
CTGGAGAAGCACTCCCTGTCTCGGGTCATCTGTGACAACACTGGCCTCACCAGGGTGCCCATGGATGCCTTCCAAGTCGGCAAATTCCCCGAAGACTTTGAGTCTTGTGACA
GCATCACTGGCATGAACCTGGAGGCCTGGAGGGAAACCTTTCCTCAAGACGACAAGTGTGGCTTCCCAGAGAGCGTGGAGAATGGGGACTTTGTGCACTGTGAGGAGTCTG
GGAGGCGCGTGCTGGTGTATTCCTGCCGGCACGGGTATGAGCTCCAAGGCCGGGAGCAGCTCACTTGCACCCAGGAAGGATGGGATTTCCAGCCTCCCCTCTGCAAAGATG
TGAACGAGTGTGCAGACGGTGCCCACCCCCCCTGCCACGCCTCTGCGAGGTGCAGAAACACCAAAGGCGGCTTCCAGTGTCTCTGCGCGGACCCCTACGAGTTAGGAGACG
ATGGGAGAACCTGCGTAGACTCCGGGAGGCTCCCTCGGGTGACTTGGATCTCCATGTCGCTGGCTGCTCTGCTGATCGGAGGCTTCGCAGGTCTCACCTCGACGGTGATTT
GCAGGTGGACACGCACTGGCACTAAATCCACACTGCCCATCTCGGAGACAGGCGGAGGAACTCCCGAGCTGAGATGCGGAAAGCACCAGGCCGTAGGGACCTCACCGCAG
CGGGCCGCAGCTCAGGACTCGGAGCAGGAGAGTGCTGGGATGGAAGGCCGGGATACTCACAGGCTGCCGAGAGCCCTCCATCATCATCATCATCATTGA

Optimized Sequence (Optimized Sequence Length: 2766bp, GC%:52.93%):

TTTTTCCCATTTATATCCCGCGGAAAGGAGCTCCTTTGGGGGAAACCTGAGGAGTCCCGCGTCTCTTCCGTCTTAGAAGAGAGCAAAAGACTCGTGGATACGGCGATGTACGC
TACCATGCAGCGTAATCTGAAAAAGAGAGGCATCTTGTCCCCTGCACAGCTTCTGAGCTTTAGTAAACTTCCCGAGCCTACAAGCGGCGTTATTGCGCGCGCAGCCGAAATTA
TGGAAACCAGTATCCAAGCTATGAAAAGGAAGGTGAACCTGAAAACGCAACAAAGTCAGCACCCCACAGACGCCCTATCAGAAGACCTCTTGAGTATTATCGCAAATATGTCG
GGCTGCCTGCCGTATATGCTCCCACCGAAATGTCCCAACACCTGCTTAGCAAATAAGTATCGGCCAATTACAGGAGCTTGTAACAACCGTGACCATCCTCGGTGGGGCGCATC
AAACACCGCGCTGGCACGGTGGCTACCTCCAGTATATGAGGATGGGTTTAGCCAGCCGAGAGGCTGGAACCCCGGGTTCTTATATAATGGATTCCCGTTACCACCCGTTAGG
GAGGTTACCAGACACGTTATACAAGTGTCTAATGAGGTAGTAACTGATGATGACCGCTACTCGGACCTGCTTATGGCTTGGGGGCAATATATAGATCACGACATTGCTTTCACG
CCACAGTCTACATCAAAAGCCGCCTTCGGAGGGGGTGCTGATTGCCAGATGACTTGCGAAAATCAGAACCCCTGCTTCCCAATTCAACTGCCCGAAGAAGCACGGCCAGCGG
CAGGAACCGCGTGTCTACCTTTTTATAGGTCGTCCGCGGCGTGCGGCACTGGGGATCAAGGCGCCCTATTTGGTAATTTGAGCACGGCAAACCCACGTCAGCAAATGAATGG
TCTTACCAGTTTTTTGGATGCGTCAACTGTTTATGGGTCATCCCCCGCATTGGAGCGACAGCTACGCAATTGGACCAGTGCTGAGGGGCTGCTACGGGTGCACGCAAGGCTG
CGAGACTCTGGGCGTGCTTATTTACCATTTGTTCCTCCGCGCGCCCCGGCCGCCTGTGCACCAGAGCCTGGTATACCGGGCGAGACTCGCGGTCCATGTTTTCTTGCGGGG
GACGGGAGGGCCTCGGAGGTTCCTAGCCTGACGGCTCTCCACACCCTATGGTTGCGAGAACATAATAGGCTGGCTGCTGCTCTCAAGGCCTTAAACGCTCATTGGAGTGCAG
ATGCCGTATACCAAGAAGCACGTAAGGTGGTAGGAGCCCTTCATCAGATAATTACTTTACGGGACTACATACCGAGAATCCTTGGCCCAGAAGCCTTTCAACAATACGTGGGA
CCGTATGAAGGGTACGACAGCACTGCTAACCCGACTGTGTCTAACGTTTTCTCGACGGCCGCATTCAGATTCGGTCATGCCACTATTCACCCTCTCGTTCGTCGACTAGATGC
CTCCTTTCAAGAACATCCAGACCTACCGGGCTTATGGTTACACCAGGCGTTTTTTTCGCCGTGGACATTACTCAGGGGAGGTGGCTTAGATCCTTTGATACGTGGATTGCTGG
CGCGCCCAGCCAAGTTGCAGGTACAGGACCAGTTAATGAACGAAGAACTTACCGAGAGGCTCTTCGTCTTGTCTAATTCTTCCACCCTCGATTTAGCTTCGATAAACCTACAGC
GGGGTAGAGATCATGGCTTGCCAGGTTACAACGAGTGGCGTGAATTTTGTGGACTCCCTCGACTTGAGACTCCGGCGGATCTGTCAACTGCGATTGCTTCTAGGAGCGTAGC
AGATAAAATCCTAGATCTCTACAAACATCCTGACAATATTGACGTATGGTTGGGTGGTCTCGCTGAAAATTTTTTACCCCGAGCACGGACAGGACCCTTGTTCGCTTGCCTCAT
CGGTAAGCAAATGAAGGCCCTTCGCGACGGTGATTGGTTTTGGTGGGAAAATTCTCATGTCTTTACCGATGCGCAACGACGGGAACTGGAAAAGCACTCCCTGTCGAGAGTA
ATATGTGACAACACCGGACTTACAAGAGTTCCCATGGACGCATTTCAAGTGGGGAAGTTTCCCGAGGATTTCGAATCCTGTGACTCTATAACAGGTATGAATCTTGAGGCGTG
GCGGGAGACATTCCCACAAGATGATAAATGCGGGTTCCCCGAGTCAGTGGAGAACGGAGACTTCGTCCATTGTGAGGAGTCGGGGAGGCGTGTGCTAGTCTACTCGTGCAG
ACATGGTTATGAATTACAGGGTCGTGAACAGCTCACGTGTACACAGGAAGGATGGGATTTTCAGCCCCCTTTATGCAAAGATGTAAATGAGTGTGCAGATGGCGCTCACCCCC
CTTGCCACGCTAGTGCACGTTGTCGCAATACTAAAGGGGGCTTCCAATGTCTGTGTGCCGACCCGTACGAACTAGGTGACGACGGTCGCACATGCGTCGACTCTGGCCGTTT
GCCGCGAGTCACGTGGATCAGTATGTCGCTTGCGGCGTTGCTGATCGGGGGCTTCGCAGGGCTAACTTCAACGGTCATCTGCCGGTGGACCCGGACGGGAACTAAGTCCAC
GCTCCCGATCTCTGAGACGGGTGGCGGTACACCCGAGTTGAGATGTGGGAAGCACCAAGCTGTCGGTACAAGCCCACAACGAGCAGCGGCTCAGGACTCAGAACAAGAAAG
CGCTGGAATGGAAGGGCGAGATACGCACCGACTTCCTCGAGCACTACACCATCACCATCACCATTGA

GC Content Adjustment:

GC Content Adjustment

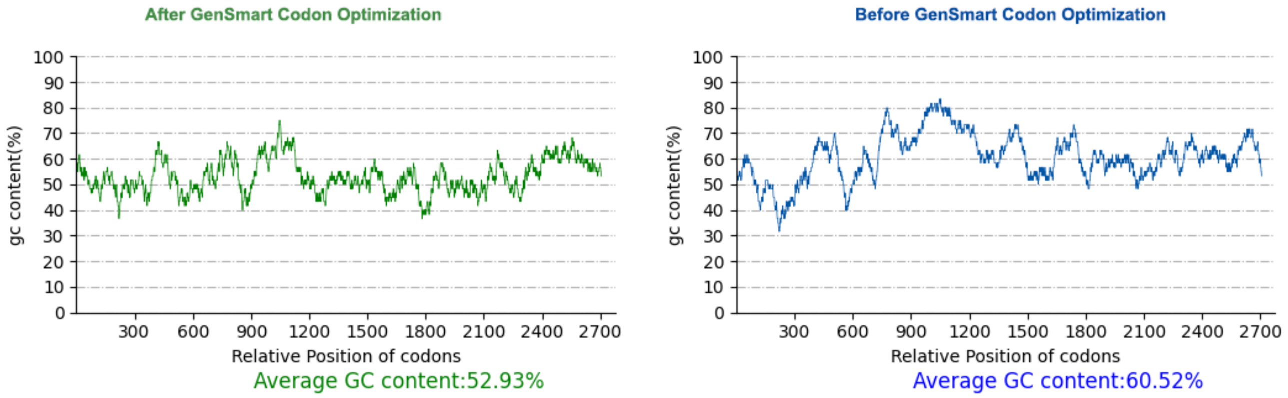

Figure 1: GC content of the sequence before and after GenSmart Codon Optimization. The ideal GC content range is between 30% and 70%.

Restriction Enzymes	Optimized	Original
EcoRI[GAATTC]	0	0
NotI[GCGGCCGC]	0	0

DNA Alignment (Optimized Region)

Optimized	1	TTTTTCCCATTTATATCCCGCGGAAAGGAGCTCCTTTGGGGGAAACCTGAGGAGTCCCGC
Original	1	TTCTTCCCCTTCATCTCGAGAGGGAAAGAACTCCTTTGGGGAAAGCCTGAGGAGTCTCGT
Optimized	61	GTCTCTTCCGTCTTAGAAGAGAGCAAAAGACTCGTGGATACGGCGATGTACGCTACCATG
Original	61	GTCTCTAGCGTCTTGGAGGAAAGCAAGCGCCTGGTGGACACCGCCATGTACGCCACGATG
Optimized	121	CAGCGTAATCTGAAAAAGAGAGGCATCTTGTCCCCTGCACAGCTTCTGAGCTTTAGTAAA
Original	121	CAGAGAAACCTCAAGAAAAGAGGAATCCTTTCTCCAGCTCAGCTTCTGTCTTTTTCCAAA
Optimized	181	CTTCCCGAGCCTACAAGCGGCGTTATTGCGCGCGCAGCCGAAATTATGGAAACCAGTATC
Original	181	CTTCCTGAGCCAACAAGCGGAGTGATTGCCCGAGCAGCAGAGATAATGGAAACATCAATA
Optimized	241	CAAGCTATGAAAAGGAAGGTGAACCTGAAAACGCAACAAAGTCAGCACCCCACAGACGCC
Original	241	CAAGCGATGAAAAGAAAAGTCAACCTGAAAACTCAACAATCACAGCATCCAACGGATGCT
Optimized	301	CTATCAGAAGACCTCTTGAGTATTATCGCAAATATGTCGGGCTGCCTGCCGTATATGCTC
Original	301	TTATCAGAAGATCTGCTGAGCATCATTGCAAACATGTCTGGATGTCTCCCTTACATGCTG
Optimized	361	CCACCGAAATGTCCCAACACCTGCTTAGCAAATAAGTATCGGCCAATTACAGGAGCTTGT
Original	361	CCCCCAAAATGCCCAAACACTTGCCTGGCGAACAAATACAGGCCCATCACAGGAGCTTGC
Optimized	421	AACAACCGTGACCATCCTCGGTGGGGCGCATCAAACACCGCGCTGGCACGGTGGCTACCT
Original	421	AACAACAGAGACCACCCCAGATGGGGCGCCTCCAACACGGCCCTGGCACGATGGCTCCCT
Optimized	481	CCAGTATATGAGGATGGGTTTAGCCAGCCGAGAGGCTGGAACCCCGGGTTCTTATATAAT
Original	481	CCAGTCTATGAGGACGGCTTCAGTCAGCCCCGAGGCTGGAACCCCGGCTTCTTGTACAAC
Optimized	541	GGATTCCCGTTACCACCCGTTAGGGAGGTTACCAGACACGTTATACAAGTGTCTAATGAG
Original	541	GGGTTCCCACTGCCCCCGGTCCGGGAGGTGACAAGACATGTCATTCAAGTTTCAAATGAG
Optimized	601	GTAGTAACTGATGATGACCGCTACTCGGACCTGCTTATGGCTTGGGGGCAATATATAGAT
Original	601	GTTGTCACAGATGATGACCGCTATTCTGACCTCCTGATGGCATGGGGACAATACATCGAC
Optimized	661	CACGACATTGCTTTCACGCCACAGTCTACATCAAAAGCCGCCTTCGGAGGGGGTGCTGAT
Original	661	CACGACATCGCGTTCACACCACAGAGCACCAGCAAAGCTGCCTTCGGGGGAGGGGCTGAC
Optimized	721	TGCCAGATGACTTGCGAAAATCAGAACCCCTGCTTCCCAATTCAACTGCCCGAAGAAGCA
Original	721	TGCCAGATGACTTGTGAGAACCAAAACCCATGTTTTCCCATACAACTCCCGGAGGAGGCC
Optimized	781	CGGCCAGCGGCAGGAACCGCGTGTCTACCTTTTTATAGGTCGTCCGCGGCGTGCGGCACT
Original	781	CGGCCGGCCGCGGGCACCGCCTGTCTGCCCTTCTACCGCTCTTCGGCCGCCTGCGGCACC

Optimized	841	GGGGATCAAGGCGCCCTATTTGGTAATTTGAGCACGGCAAACCCACGTCAGCAAATGAAT
Original	841	GGGGACCAAGGCGCGCTCTTTGGGAACCTGTCCACGGCCAACCCGCGGCAGCAGATGAAC
Optimized	901	GGTCTTACCAGTTTTTTGGATGCGTCAACTGTTTATGGGTCATCCCCCGCATTGGAGCGA
Original	901	GGGTTGACCTCGTTCCTGGACGCGTCCACCGTGTATGGCAGCTCCCCGGCCCTAGAGAGG
Optimized	961	CAGCTACGCAATTGGACCAGTGCTGAGGGGCTGCTACGGGTGCACGCAAGGCTGCGAGAC
Original	961	CAGCTGCGGAACTGGACCAGTGCCGAAGGGCTGCTCCGCGTCCACGCGCGCCTCCGGGAC
Optimized	1021	TCTGGGCGTGCTTATTTACCATTTGTTCCTCCGCGCGCCCCGGCCGCCTGTGCACCAGAG
Original	1021	TCCGGCCGCGCCTACCTGCCCTTCGTGCCGCCACGCGCGCCTGCGGCCTGTGCGCCCGAG
Optimized	1081	CCTGGTATACCGGGCGAGACTCGCGGTCCATGTTTTCTTGCGGGGGACGGGAGGGCCTCG
Original	1081	CCCGGCATCCCCGGAGAGACCCGCGGGCCCTGCTTCCTGGCCGGAGACGGCCGCGCCAGC
Optimized	1141	GAGGTTCCTAGCCTGACGGCTCTCCACACCCTATGGTTGCGAGAACATAATAGGCTGGCT
Original	1141	GAGGTCCCCTCCCTGACGGCACTGCACACGCTGTGGCTGCGCGAGCACAACCGCCTGGCC
Optimized	1201	GCTGCTCTCAAGGCCTTAAACGCTCATTGGAGTGCAGATGCCGTATACCAAGAAGCACGT
Original	1201	GCGGCGCTCAAGGCCCTCAATGCGCACTGGAGCGCGGACGCCGTGTACCAGGAGGCGCGC
Optimized	1261	AAGGTGGTAGGAGCCCTTCATCAGATAATTACTTTACGGGACTACATACCGAGAATCCTT
Original	1261	AAGGTCGTGGGCGCTCTGCACCAGATCATCACCCTGAGGGATTACATCCCCAGGATCCTG
Optimized	1321	GGCCCAGAAGCCTTTCAACAATACGTGGGACCGTATGAAGGGTACGACAGCACTGCTAAC
Original	1321	GGACCCGAGGCCTTCCAGCAGTACGTGGGTCCCTATGAAGGCTATGACTCCACCGCCAAC
Optimized	1381	CCGACTGTGTCTAACGTTTTCTCGACGGCCGCATTCAGATTCGGTCATGCCACTATTCAC
Original	1381	CCCACTGTGTCCAACGTGTTCTCCACAGCCGCCTTCCGCTTCGGCCATGCCACGATCCAC
Optimized	1441	CCTCTCGTTCGTCGACTAGATGCCTCCTTTCAAGAACATCCAGACCTACCGGGCTTATGG
Original	1441	CCGCTGGTGAGGAGGCTGGACGCCAGCTTCCAGGAGCACCCCGACCTGCCCGGGCTGTGG
Optimized	1501	TTACACCAGGCGTTTTTTTCGCCGTGGACATTACTCAGGGGAGGTGGCTTAGATCCTTTG
Original	1501	CTGCACCAGGCTTTCTTCAGCCCATGGACATTACTCCGTGGAGGTGGTTTGGACCCACTA
Optimized	1561	ATACGTGGATTGCTGGCGCGCCCAGCCAAGTTGCAGGTACAGGACCAGTTAATGAACGAA
Original	1561	ATACGAGGCCTTCTTGCAAGACCAGCCAAACTGCAGGTGCAGGATCAGCTGATGAACGAG
Optimized	1621	GAACTTACCGAGAGGCTCTTCGTCTTGTCTAATTCTTCCACCCTCGATTTAGCTTCGATA
Original	1621	GAGCTGACGGAAAGGCTCTTTGTGCTGTCCAATTCCAGCACCTTGGATCTGGCGTCCATC
Optimized	1681	AACCTACAGCGGGGTAGAGATCATGGCTTGCCAGGTTACAACGAGTGGCGTGAATTTTGT
Original	1681	AACCTGCAGAGGGGCCGGGACCACGGGCTGCCAGGTTACAATGAGTGGAGGGAGTTCTGC
Optimized	1741	GGACTCCCTCGACTTGAGACTCCGGCGGATCTGTCAACTGCGATTGCTTCTAGGAGCGTA
Original	1741	GGCCTGCCTCGCCTGGAGACCCCCGCTGACCTGAGCACAGCCATCGCCAGCAGGAGCGTG
Optimized	1801	GCAGATAAAATCCTAGATCTCTACAAACATCCTGACAATATTGACGTATGGTTGGGTGGT
Original	1801	GCCGACAAGATCCTGGACTTGTACAAGCATCCTGACAACATCGATGTCTGGCTGGGAGGC
Optimized	1861	CTCGCTGAAAATTTTTTACCCCGAGCACGGACAGGACCCTTGTTCGCTTGCCTCATCGGT
Original	1861	TTAGCTGAAAACTTCCTCCCCAGGGCTCGGACAGGGCCCCTGTTTGCCTGTCTCATTGGG
Optimized	1921	AAGCAAATGAAGGCCCTTCGCGACGGTGATTGGTTTTGGTGGGAAAATTCTCATGTCTTT
Original	1921	AAGCAGATGAAGGCTCTGCGGGACGGTGACTGGTTTTGGTGGGAGAACAGCCACGTCTTC
Optimized	1981	ACCGATGCGCAACGACGGGAACTGGAAAAGCACTCCCTGTCGAGAGTAATATGTGACAAC
Original	1981	ACGGATGCACAGAGGCGTGAGCTGGAGAAGCACTCCCTGTCTCGGGTCATCTGTGACAAC
Optimized	2041	ACCGGACTTACAAGAGTTCCCATGGACGCATTTCAAGTGGGGAAGTTTCCCGAGGATTTC
Original	2041	ACTGGCCTCACCAGGGTGCCCATGGATGCCTTCCAAGTCGGCAAATTCCCCGAAGACTTT
Optimized	2101	GAATCCTGTGACTCTATAACAGGTATGAATCTTGAGGCGTGGCGGGAGACATTCCCACAA
Original	2101	GAGTCTTGTGACAGCATCACTGGCATGAACCTGGAGGCCTGGAGGGAAACCTTTCCTCAA
Optimized	2161	GATGATAAATGCGGGTTCCCCGAGTCAGTGGAGAACGGAGACTTCGTCCATTGTGAGGAG
Original	2161	GACGACAAGTGTGGCTTCCCAGAGAGCGTGGAGAATGGGGACTTTGTGCACTGTGAGGAG
Optimized	2221	TCGGGGAGGCGTGTGCTAGTCTACTCGTGCAGACATGGTTATGAATTACAGGGTCGTGAA
Original	2221	TCTGGGAGGCGCGTGCTGGTGTATTCCTGCCGGCACGGGTATGAGCTCCAAGGCCGGGAG
Optimized	2281	CAGCTCACGTGTACACAGGAAGGATGGGATTTTCAGCCCCCTTTATGCAAAGATGTAAAT

Original	2281	CAGCTCACTTGCACCCAGGAAGGATGGGATTTCCAGCCTCCCCTCTGCAAAGATGTGAAC
Optimized	2341	GAGTGTGCAGATGGCGCTCACCCCCCTTGCCACGCTAGTGCACGTTGTCGCAATACTAAA
Original	2341	GAGTGTGCAGACGGTGCCCACCCCCCCTGCCACGCCTCTGCGAGGTGCAGAAACACCAAA
Optimized	2401	GGGGGCTTCCAATGTCTGTGTGCCGACCCGTACGAACTAGGTGACGACGGTCGCACATGC
Original	2401	GGCGGCTTCCAGTGTCTCTGCGCGGACCCCTACGAGTTAGGAGACGATGGGAGAACCTGC
Optimized	2461	GTCGACTCTGGCCGTTTGCCGCGAGTCACGTGGATCAGTATGTCGCTTGCGGCGTTGCTG
Original	2461	GTAGACTCCGGGAGGCTCCCTCGGGTGACTTGGATCTCCATGTCGCTGGCTGCTCTGCTG
Optimized	2521	ATCGGGGGCTTCGCAGGGCTAACTTCAACGGTCATCTGCCGGTGGACCCGGACGGGAACT
Original	2521	ATCGGAGGCTTCGCAGGTCTCACCTCGACGGTGATTTGCAGGTGGACACGCACTGGCACT
Optimized	2581	AAGTCCACGCTCCCGATCTCTGAGACGGGTGGCGGTACACCCGAGTTGAGATGTGGGAAG
Original	2581	AAATCCACACTGCCCATCTCGGAGACAGGCGGAGGAACTCCCGAGCTGAGATGCGGAAAG
Optimized	2641	CACCAAGCTGTCGGTACAAGCCCACAACGAGCAGCGGCTCAGGACTCAGAACAAGAAAGC
Original	2641	CACCAGGCCGTAGGGACCTCACCGCAGCGGGCCGCAGCTCAGGACTCGGAGCAGGAGAGT
Optimized	2701	GCTGGAATGGAAGGGCGAGATACGCACCGACTTCCTCGAGCACTACACCATCACCATCAC
Original	2701	GCTGGGATGGAAGGCCGGGATACTCACAGGCTGCCGAGAGCCCTCCATCATCATCATCAT
Optimized	2761	CATTGA
Original	2761	CATTGA

Citation

If you would like to cite the GenSmart™ Codon Optimization Tool, please include the following information:

The sequence was codon-optimized using the GenSmart™ Codon Optimization Tool (https://www.genscript.com/tools/gensmart-codon-optimization) [1].

1. Long Fan (2020, February 6). *Codon optimization*. (WO Patent WO 2020/024917 A1). Nanjing GenScript Biotech Co., Ltd.